INTRODUCTION:

The RISC-V Processor Architecture is an important modern processor architecture based on the Reduced Instruction Set Computer (RISC) concept. It provides a simple, efficient, and flexible way of designing processors. RISC-V is an open standard instruction set architecture (ISA), which means that it can be used and modified by researchers, students, companies, and hardware designers without depending on a proprietary instruction set

A processor is made up of several important components such as logic gates, registers, multiplexers, arithmetic logic units (ALU), control units, memory units, and buses. These digital components work together to execute instructions. Digital logic circuits form the basic building blocks of a processor, while the instruction set provides the commands that tell the processor what operations to perform.

RISC-V connects these two levels by providing a clear relationship between digital hardware and instruction execution. When a RISC-V instruction is given to a processor, the instruction is first fetched from memory and then decoded by the control unit. Based on the instruction, different hardware components such as registers and the ALU are activated to perform the required operation.

The architecture follows a load-store design, where arithmetic and logical operations are mainly performed on data stored in registers. Data is transferred between memory and registers using specific load and store instructions. This simple approach helps in designing processors that are efficient and easier to understand.

One of the important features of RISC-V is its modular architecture. A basic RISC-V processor can be extended with additional instruction-set extensions depending on the required application. For example, extensions are available for multiplication and division, atomic operations, floating-point calculations, and other specialized functions.

Therefore, the study of "RISC-V Processor Architecture: From Digital Logic Circuits to Instruction Execution" provides a practical understanding of how a modern processor is designed and how instructions are executed inside it. It shows the connection between basic digital circuits and higher-level computer architecture, making RISC-V a useful architecture for understanding the internal working of processors.

This case study also helps demonstrate how basic concepts of Digital Principles and Computer Organization are connected in a real processor architecture. Concepts choplogic gates, combinational circuits, sequential circuits, registers

Background:

A processor is the main part of a computer system that performs calculations, processes data, and executes instructions. The working of a processor is based on digital logic circuits such as logic gates, multiplexers, registers, adders, and flip-flops. These circuits work together to perform basic operations and control the movement of data inside the processor. Understanding these digital circuits is important for understanding how a computer executes instructions.

RISC-V is an open-standard Instruction Set Architecture (ISA) based on the Reduced Instruction Set Computer (RISC) concept. It provides a simple and flexible set of instructions that can be used to design different types of processors. Unlike proprietary processor architectures, RISC-V is openly available, which makes it useful for education, research, embedded systems, and processor development.

The RISC-V processor connects the concepts of digital logic and computer architecture. When a RISC-V instruction is given to the processor, it passes through different stages before the required operation is completed. The instruction is first fetched from memory and then decoded to understand what operation has to be performed. After decoding, the required data is obtained from registers, and the Arithmetic Logic Unit (ALU) performs arithmetic or logical operations.

Digital logic circuits play an important role in each stage of instruction execution. Registers store temporary data and instructions, multiplexers select the required data paths, adders perform addition, and control circuits generate signals that coordinate the processor's activities. The ALU uses logic gates and other digital circuits to perform operations such as addition, subtraction, comparison, and logical operations.

The study of “RISC-V Processor Architecture: From Digital Logic Circuits to Instruction Execution” helps to understand how simple electronic circuits can be combined to form a complete processor. It explains the relationship between hardware components, instruction formats, registers, memory, control units, and the ALU. By studying this architecture, students can understand how an instruction written in a program is finally converted into operations performed by hardware.

RISC-V also provides a clear platform for learning processor design because its architecture is simple, modular, and easy to understand. Therefore, studying RISC-V from the level of digital logic circuits to instruction execution gives a complete view of how modern processors work and how hardware and software interact with each other

Problem Statement:

Modern computer systems depend on processors to perform calculations, process information, and execute program instructions. However, understanding how a processor actually works at the hardware level can be difficult for beginners. A processor is not a single component; it is made up of several digital circuits and functional units that work together to perform each instruction. Therefore, there is a need to understand the connection between digital logic circuits, processor architecture, and instruction execution in a simple and systematic way.

The RISC-V processor architecture provides a useful platform for studying this connection. RISC-V is based on the Reduced Instruction Set Computer (RISC) concept and uses a simple set of instructions. To execute an instruction, the processor must perform several operations such as fetching the instruction from memory, decoding the instruction, reading data from registers, performing the required operation, and storing the result. Each of these operations depends on different hardware components working together correctly.

The main problem addressed in this case study is to understand how basic digital logic circuits are combined to form the major components of a RISC-V processor and how these components work together during instruction execution. Components such as logic gates, registers, multiplexers, adders, the Arithmetic Logic Unit (ALU), control unit, and memory are essential for the proper functioning of the processor. Understanding their individual roles as well as their interaction is important for learning computer organization and processor design.

Another important aspect is understanding how a RISC-V instruction moves through the processor. The instruction must be fetched, interpreted, and executed according to its instruction format. The control unit generates appropriate control signals, while the data path transfers data between registers, the ALU, and memory. A small error in the control or data path can affect the execution of the instruction

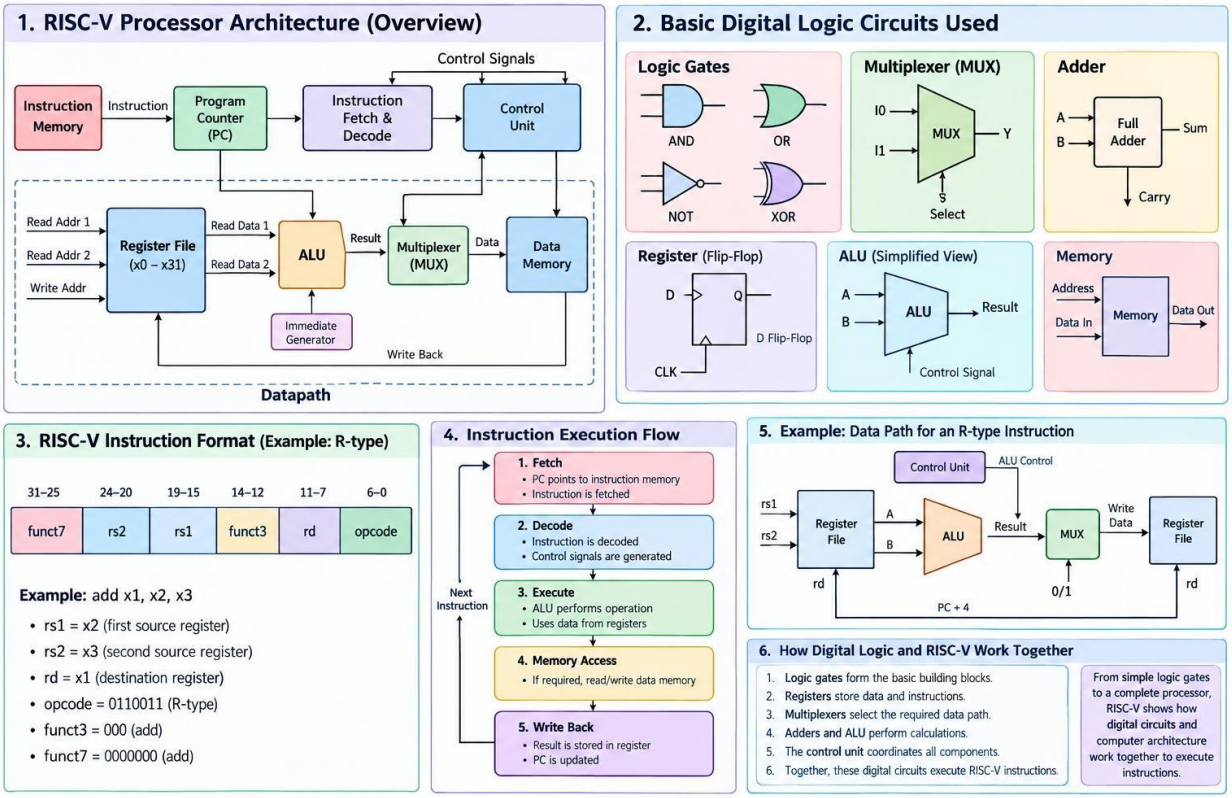
Therefore, this case study focuses on explaining the complete path from digital logic circuits to RISC-V instruction execution. It aims to provide a clear understanding of how simple hardware components are combined to create a functional processor and how software instructions are converted into operations performed by digital hardware. This knowledge helps students understand the basic principles of processor design and the interaction between hardware and software.

Analysis:

The analysis of the RISC-V Processor Architecture focuses on understanding how different digital logic components work together to execute an instruction. A RISC-V processor is designed using a combination of a data path and control unit, where each component performs a specific function. The major components include registers, Arithmetic Logic Unit (ALU), multiplexers, adders, memory, and control circuits.

The first important part is the register file, which stores temporary data and operands required during instruction execution. RISC-V provides a set of general-purpose registers that can be accessed by instructions. The register file allows the processor to quickly read required data and store the results of operations

The ALU is another major component of the processor. It performs arithmetic and logical operations such as addition, subtraction, AND, OR, and comparisons. The ALU is constructed using digital logic circuits such as logic gates, adders, and multiplexers. The operation performed by the ALU is determined by control signals generated by the control unit.



Thus, the analysis shows that RISC-V provides a clear connection between digital logic, computer organization, and instruction execution. Understanding this relationship helps explain how a simple instruction is converted into a sequence of hardware operations and how different processor components work together to complete the execution process.

Digital logic circuits form the foundation of these operations. Logic gates perform basic logical functions, flip-flops and registers store information, multiplexers select data, and adders perform arithmetic operations. These circuits are connected together to create the ALU, control circuits, and other processor.

Example: RISC-V ADD Instruction:

Consider the instruction:

ADD x5, x6, x7

This instruction means that the values stored in registers x6 and x7 are added, and the result is stored in x5.

Execution steps:

1) Fetch: The processor fetches the ADD instruction from memory.
2) Decode: The control unit identifies it as an ADD operation.
3) Read Registers: Values from x6 and x7 are read.
4) ALU Operation: The ALU adds the two values.

Write Back: The result is stored in x5.

For example, if:

x6 = 10

x7 = 20

Then:

x5 = x6 + x7

x5 = 10 + 20 = 30

So, the final value stored in x5 is 30.

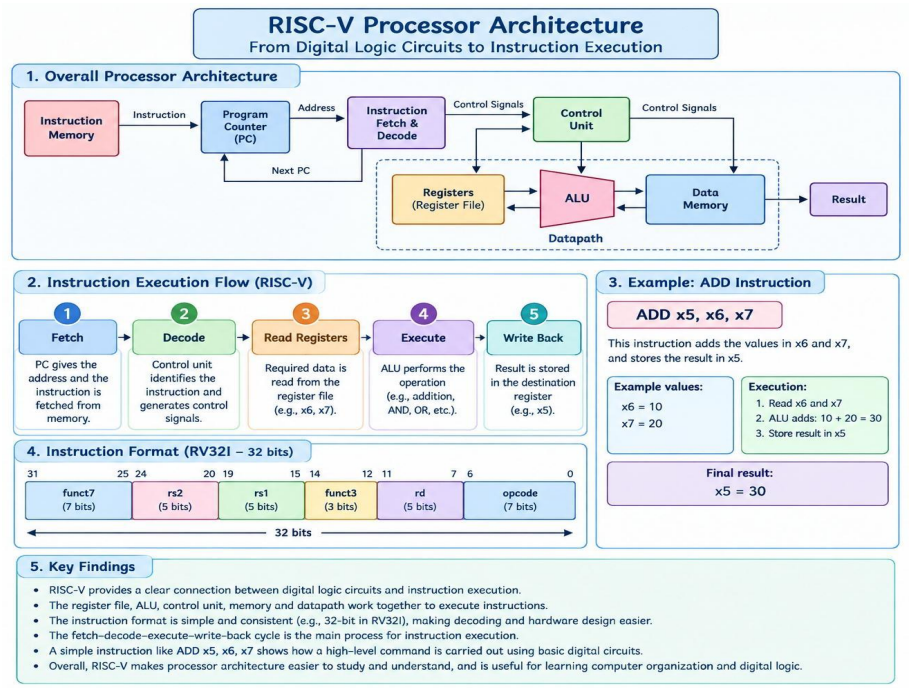
Findings:

The study found that RISC-V provides a clear connection between digital logic circuits and instruction execution. RISC-V is an open-standard Instruction Set Architecture (ISA), with a base instruction set that can be extended for different processor designs.

One important finding is that the registers, ALU, control unit, memory and datapath work together to execute instructions. The Program Counter helps identify the instruction to be fetched, while registers hold the data required during execution. This shows how processor instructions are supported by basic digital hardware.

The study also found that RISC-V uses a simple and organized instruction format. In the RV32I base ISA, instructions are normally 32 bits long, and the positions of source and destination registers are kept consistent to simplify instruction decoding and hardware design.

Diagram:



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Example – RISC-V Instruction Execution:

A simple example of RISC-V instruction execution is the addition of two numbers. Suppose register x1 contains 10 and register x2 contains 20. We want to add these two values and store the result in register x3.

The RISC-V instruction used is:

ADD x3, x1, x2

First, the processor fetches this instruction from memory. The Program Counter (PC) gives the address of the instruction. The instruction is then brought into the processor for processing.

Next, the processor decodes the instruction. The control unit identifies that the instruction is an ADD operation and that the values must be taken from registers x1 and x2. The register file provides the values 10 and 20.

Then, the two values are sent to the Arithmetic Logic Unit (ALU). The ALU performs the addition:

10 + 20 = 30

The result 30 is then sent back to the register file and stored in register x3.

This process can be represented as:

Memory → Instruction Fetch → Decode → Register Read → ALU → Register x3

In this example, digital logic circuits such as registers, multiplexers, control circuits and the ALU work together to execute one RISC-V instruction. Thus, the example clearly shows how a simple instruction is converted into hardware operations inside the processor.

Result:

x1 = 10, x2 = 20 → ADD → x3 = 30

This example demonstrates the connection between digital logic circuits and instruction execution in a RISC-V processor.

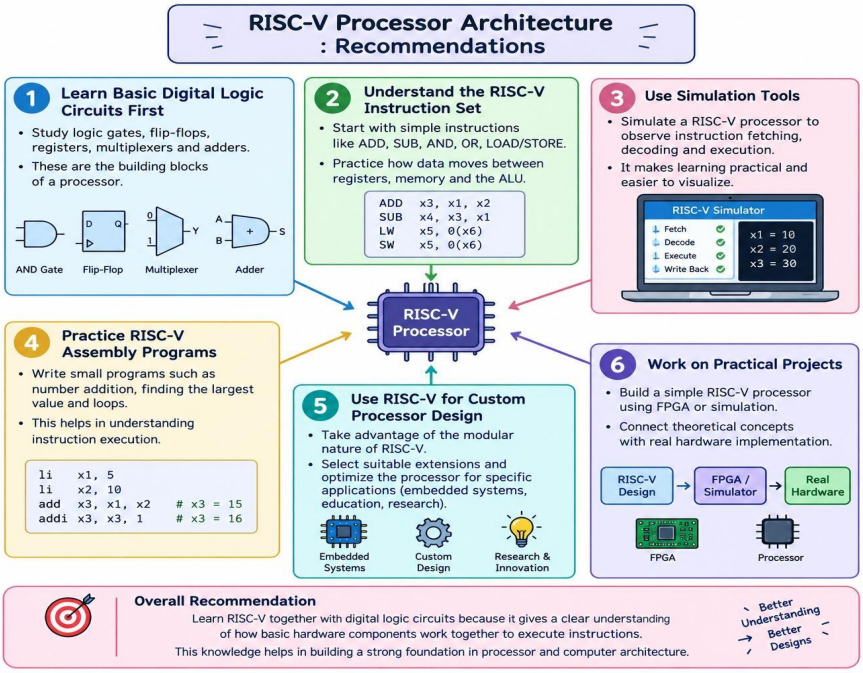
Recommendation:

The study of RISC-V Processor Architecture: From Digital Logic Circuits to Instruction Execution provides several useful recommendations for understanding and developing processor systems. First, students should learn the basic concepts of digital logic circuits, such as logic gates, flip-flops, registers, multiplexers and adders, before studying processor architecture. These concepts help in understanding how different processor components work together.

It is recommended to understand the RISC-V instruction set step by step. Students can begin with simple instructions such as ADD, SUB, AND, OR and LOAD/STORE instructions. Practicing these instructions helps students understand how data moves between registers, memory and the ALU.

Another recommendation is to use simulation tools to observe processor operations. Simulating a RISC-V processor can help students understand instruction fetching, decoding and execution without requiring physical hardware. It also makes learning more practical and easier to visualize.

Students should also practice writing small RISC-V assembly programs. Simple programs such as adding numbers, finding the largest value and performing loops can help develop a better understanding



Examples:

A practical example of RISC-V processor architecture can be understood through the execution of a simple addition instruction. Consider two values, 10 and 20, stored in registers x1 and x2. The processor has to add these values and store the result in x3.

The instruction can be represented as:

ADD x3, x1, x2

This means that the processor should add the values in x1 and x2 and place the answer in x3.

1. Instruction Fetch

First, the Program Counter (PC) gives the address of the required instruction. The instruction is fetched from the instruction memory. The PC is then updated so that the processor can access the next instruction.

2. Instruction Decode

The fetched instruction is sent to the Control Unit. The processor identifies that the instruction is an ADD operation. It also identifies x1 and x2 as the source registers and x3 as the destination register.

3. Register Read

The register file provides the values stored in x1 and x2.

x1 = 10

x2 = 20

These two values are sent to the ALU for processing.

4. ALU Operation

The Arithmetic Logic Unit (ALU) performs the required addition operation.

10 + 20 = 30

The ALU produces 30 as the output.

5. Write Back

The result from the ALU is sent back to the register file. The value 30 is stored in the destination register x3.

x3 = 30

6. Role of Digital Logic Circuits

This complete operation is possible because of several digital logic components. Logic gates perform basic logical operations, multiplexers select the required data, registers store values, and adders perform arithmetic operations. The control unit coordinates these components.

Example Flow

Instruction Memory → Instruction Decode → Register File → ALU → Register File

ADD x3, x1, x2

10 + 20 = 30

Result: x3 = 30

This example clearly shows how a simple RISC-V instruction moves through different processor components. It also demonstrates the connection between digital logic circuits and instruction execution. Thus, RISC-V provides a simple and systematic way to understand how a processor executes instructions from beginning to end.

Conclusion :

The study of "RISC-V Processor Architecture: From Digital Logic Circuits to Instruction Execution" provides a clear understanding of how a processor works from the basic hardware level to the execution of instructions. RISC-V is a simple and flexible instruction set architecture that helps in understanding the internal working of modern processors.

The study shows that digital logic circuits are the foundation of processor design. Components such as logic gates, registers, multiplexers, adders, ALU, control unit, and memory work together to perform different operations. Each component has a specific role in processing and controlling data.

RISC-V follows a systematic instruction execution process. An instruction is first fetched from memory, then decoded by the control unit. The required data is obtained from registers and processed by the ALU. Finally, the result is stored in the required register or memory location. This process helps the processor execute programs correctly and efficiently.

The example of the ADD instruction shows how these components work together. When two values are stored in registers, the ALU performs the addition and the result is written back to the destination register. This demonstrates the connection between basic digital circuits and actual instruction execution.

Another important conclusion is that RISC-V has a modular and flexible architecture. Its open instruction set allows researchers, students, and developers to study processor design and create customized processor systems. It can be useful in education, embedded systems, research, and different computing applications.

Overall, RISC-V provides a strong foundation for understanding computer organization and processor architecture. The study connects basic digital logic concepts with practical instruction execution. Therefore, learning RISC-V helps students understand how software instructions are converted into hardware operations and how different processor components work together to complete a task.

Example:

A simple example of RISC-V instruction execution is adding two numbers. Suppose the processor has two values, 10 and 20, stored in registers. The RISC-V processor receives an addition instruction to add these values.

First, the processor fetches the instruction from memory. Then the control unit decodes the instruction and identifies the required registers. The two values are sent to the ALU (Arithmetic Logic Unit). The ALU performs the addition using digital logic circuits.

10 + 20 = 30

Finally, the result 30 is stored in the destination register. This example shows how registers, control unit, ALU, and digital logic circuits work together to execute a RISC-V instruction.

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- Technical resources on digital logic circuits, ALU, registers, memory and processor datapath.